Attack-Aware Market Making: Adversarial Co-Training with Economically-Constrained Spoofing Agents

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1 Background and Motivation

Spoofing, placing large orders with no intention of execution to artificially shift apparent supply or demand, then cancelling before fill, is illegal under the Dodd-Frank Act and has contributed to well-documented market instability events [4]. Deep reinforcement learning (DRL) market-making agents are structurally vulnerable: because they condition directly on limit order book (LOB) state [2], a large spoof shifts the apparent mid-price and causes the agent to misquote in the adversary's intended direction, persisting even in agents trained under benign conditions [5].

The gap. Spooner & Savani (2020) demonstrated improved robustness via adversarial reinforcement learning (RL) [5], but three limitations remain. First, existing adversaries are unconstrained (no fees, capital costs, or fill penalties), overstating attack severity and potentially hardening defenders against economically impossible threats. Second, no prior work equips the market maker with a mechanism to detect that an attack is occurring in real time; defence is purely reactive, with no signal to the policy that spoofed quotes should be discounted. Third, existing approaches do not condition on market regime, limiting generalisation across high- and low-volatility environments with structurally different spread dynamics and fill probabilities.

Related work. The Avellaneda-Stoikov (A-S) model [1] provides an analytically tractable closed-form benchmark for inventory-aware quoting; JaxMARL-HFT retains it as a reference policy, making it a natural performance floor for the IPPO agent. Spooner & Savani [5] remain the primary methodological baseline: they train a market maker against an unconstrained adversarial agent and demonstrate measurable robustness gains, but their formulation lacks economic constraints on the adversary, any detection mechanism, and regime conditioning, the three gaps this work directly addresses. Wang et al. [8] introduce competitive MARL market-making in which agents implicitly discipline each other's spreads, but include no adversarial spoofing agent and no detection or regime-awareness component. Wang & Wellman [6] frame spoofing detection as a standalone adversarial game between separate agents; by contrast, this work embeds detection as an auxiliary objective within the market maker's own policy network, so the detection signal is available to the policy during execution rather than only as a post-hoc label. Byrd [7] uses norm-augmented RL to suppress spoofing behaviour in a trading agent, inverting the detection framing and not addressing robustness of a separate market maker. Briola et al. [9] demonstrate that common attention-based LOB forecasting architectures admit look-ahead bias through improper temporal masking; causal masking is applied to all attention operations here as a direct safeguard. Caruana's multi-task learning framework [12] motivates the auxiliary detection head: jointly training shared representations on related objectives regularises the encoder and can improve sample efficiency, relevant given the limited number of distinct attack episodes per training run. Do & Putnins [4] provide empirical grounding for the spoofing-efficacy/regime relationship and a set of order-flow signatures (order-size asymmetry, cancellation-rate spikes, short-lived depth imbalances) that inform the feature set exposed to the detection head.

2 Research Objectives

Research question. Can adversarially co-trained market-making agents equipped with an explicit attack-detection head and a binary volatility-regime indicator maintain significantly higher risk-adjusted performance under economically-constrained spoofing attacks, and does this robustness generalise across high- and low-volatility regimes without degrading normal-market performance?

Framing. This is treated as an exploratory study. Effect sizes are reported with Cohen's d (standardised mean difference in metrics across ≈20 seeds) and 95% confidence intervals. Shapiro–Wilk tests ($\alpha = 0.05$) determine whether parametric or non-parametric fallbacks apply; G*Power targets $d \geq 0.8$ at $\alpha = 0.05$ with ≈20 seeds.

Contributions.

1. **Economically-constrained adversary (evaluation):** an unconstrained spoofing agent is used during adversarial co-training; after training, the market maker is evaluated against a budget-constrained variant whose cost function incorporates order fees (c_1), capital commitment costs (c_2), and accidental fill penalties (c_3). The instrument is AMZN (NASDAQ equity); costs are therefore calibrated against NASDAQ equity maker/taker fee schedules and per-share disgorgement figures from Securities and Exchange Commission (SEC) and Department of Justice (DOJ) equity spoofing enforcement actions — not futures-market schedules. Phase 1 will confirm the applicable fee tier and identify a suitable enforcement action for c_3. Constraining the adversary at evaluation rather than training preserves internal validity (both baseline and defended agents face the same adversary) while ensuring external validity: results reflect attack intensities a real spoofer would sustain given fees, capital costs, and disgorgement risk.
2. **Attack-detection auxiliary head:** a binary classification head jointly trained with the market maker's policy predicts whether a non-bona-fide order is active in the LOB. A search of ICAIF (2019–2025), IJCAI (2018–2025), AAAI (2017–2025), AAMAS (2018–2025), and NeurIPS FinRL proceedings found no prior work treating spoof-detection as an auxiliary objective within a DRL market maker's policy network. Wang & Wellman [6] treat detection as a standalone framework between separate agents; Byrd [7] uses RL to *suppress* spoofing in a trading agent, the inverse of detection.
3. **Volatility-regime conditioning:** a binary high/low indicator derived from 20-day rolling realised volatility of AMZN, thresholded at its trailing median, is appended to the agent's state. This requires no offline training and introduces one additional input dimension. Whether this conditioning improves robustness is treated empirically; if it does not, the result is reported as a null finding.

3 Method

3.1 Simulation Platform and Problem Formulation

Training uses JaxMARL-HFT [3] (∼240× CPU speedup) on AMZN 2024 LOBSTER data, matching Mohl et al. directly. Engineering challenges include XLA recompilation on architecture changes (addressed via a Phase 1 training-loop validation milestone), policy-freeze synchronisation in JAX's functional paradigm (requires pytree partitioning not natively supported by JaxMARL-HFT), and correct device broadcast of the detection signal and regime indicator across the vectorised batch.

The interaction is a general-sum two-player Markov game; neither agent observes the other's policy. Joint state: top-10 LOB levels, inventory, volume-weighted average price (VWAP), detection signal, and binary regime indicator. Market-replay simulation does not causally propagate injected orders: the adversary modifies *apparent* LOB state only. This is a deliberate simplification: a fully causal simulator would require modelling the market's response to spoofed volume, introducing a separate and substantial estimation problem. The replay assumption means that adversary order sizes are bounded so that injected depth does not exceed realistic proportions of observed best-quote depth at the corresponding timestamp.

The market maker's reward follows Spooner & Savani [5]:

$$R_t^{MM} = \Delta \text{PnL}_t - \phi \cdot \text{Var}(\text{PnL}_t) - \lambda |q_t|$$

where ϕ penalises profit volatility and λ penalises inventory q_t.

Non-stationarity. IPPO [10] treats the opponent as part of the environment and updates each agent's policy independently. In a two-player competitive game this introduces non-stationarity: as the adversary policy improves, the market maker's environment distribution shifts, and vice versa. Alternating policy freezes partially mitigate this by holding one agent fixed while the other adapts, but they do not eliminate the issue and can cause cycling if freeze schedules are poorly tuned. This is a known challenge in competitive MARL [10, 11]; the ablation study will include a sweep over freeze-schedule hyperparameters to assess sensitivity.

3.2 Challenge 1: Economically-Constrained Adversary

During co-training the adversary optimises $R_t^{ADV} = -R_t^{MM}$, unconstrained, to produce a maximally hard training signal. At evaluation, adversary actions are filtered through an economic feasibility check: any spoofing sequence whose cumulative cost exceeds budget B is replaced with a no-op. Per-action cost:

$$\text{cost}_t = c_1 N_{\text{orders}} + c_2 K_{\text{committed}} + c_3 P_{\text{fill}}$$

c_1: NASDAQ equity taker fee (≈\$0.003/share); c_2: overnight funding rate on committed notional; c_3: median per-share disgorgement from SEC/DOJ equity enforcement actions (confirmed Phase 1). B is a sensitivity parameter. The adversary selects: side, order size (multiple of best-quote depth), and tick placement; orders are tagged non-bona-fide internally, yielding detection labels at no extra cost. Layering extends the action space to $n \in \{1, 2, 3\}$ simultaneous orders per side. Both agents train via independent proximal policy optimisation (IPPO) [10] with alternating policy freezes, a standard but imperfect mitigation for non-stationarity in competitive MARL [11].

3.3 Challenge 2: Attack-Detection Auxiliary Head

A binary classification head on the market maker's shared encoder predicts $\hat{y}_t \in [0, 1]$, with oracle labels from simulator adversary tags. The combined training objective [12]:

$$\mathcal{L} = \mathcal{L}_{\text{PPO}} + \lambda_c \cdot \mathcal{L}_{\text{BCE}}(\hat{y}_t, y_t)$$

$\hat{y}_t$ feeds back into agent state, allowing the policy to discount apparent LOB imbalance during suspected attacks. If detection AUROC does not exceed a minimum viability threshold (established in Phase 1 baseline experiments) by mid-Phase 3, the head is dropped and this is reported. Oracle labels are unavailable in live markets; reported AUROC is an upper bound and the contribution is scoped as a simulation feasibility demonstration, not a claim of deployability to live surveillance.

3.4 Challenge 3: Volatility-Regime Conditioning

A binary indicator $z_t \in \{0, 1\}$ is computed from 20-day rolling realised volatility of AMZN thresholded at its trailing 252-day median ($z_t = 1$: high volatility). It is appended to the observation vector with no modification to the RL training loop. If the indicator does not measurably improve robustness, this is reported as a null finding.

4 Evaluation

Primary metrics: annualised Sortino and Sharpe ratios over *attack-on* windows. Sortino is the preferred robustness metric because attack windows induce negatively skewed, directionally biased returns; Sharpe's use of total volatility conflates systematic damage with symmetric noise and can appear flattering even as the agent is consistently picked off. Sharpe is retained as co-primary for comparability with Spooner & Savani [5] and prior work. SoftMin Sharpe — which downweights outlier returns via a differentiable softmin — is retained as a robustness check in the sensitivity sweep, as it targets episodic outliers rather than the sustained mean-drift spoofing produces. **Secondary:** $\mathrm{CVaR}_{0.10}$ for tail-loss exposure (10% threshold chosen because at ≈20 seeds the 5th-percentile effective sample is ∼1 observation, making $\mathrm{CVaR}_{0.05}$ statistically uninformative). **Behavioural:** *quote displacement* (mean absolute deviation of quoted price from fair value under attack vs. matched clean conditions at the same LOB state) distinguishes genuine robustness from favourable seed variance; *peak inventory excursion* during attack windows tests for the directional accumulation characteristic of a successfully spoofed agent. **Diagnostic:** detection AUROC over attack-on windows.

All metrics are computed under unconstrained and budget-constrained adversary conditions (see Section 3.2) and over *attack-off* windows. Statistical tests use paired t-tests with Cohen's d and 95% confidence intervals; Wilcoxon signed-rank tests and bootstrap confidence intervals serve as non-parametric fallbacks if Sortino or Sharpe distributions depart significantly from normality (Shapiro–Wilk, $\alpha = 0.05$, as described in Section 2). A sensitivity sweep over ϕ, λ, λ_c, and B accompanies the ablation study.

Agent configurations: (1) **Baseline**: A-S and vanilla IPPO; (2) **Adversarial IPPO**: co-trained, no detection head or regime indicator; (3) **Full model**: adversarial IPPO with detection head and regime indicator. Configs 1 vs. 2 isolate adversarial co-training; 2 vs. 3 isolate the detection and regime contributions.

Regime evaluation windows: the three longest contiguous windows ($\geq$ 20 trading days) of each regime label in the held-out data.

Progression gate: Baseline IPPO must achieve Sortino and Sharpe within an acceptable margin of A-S and inventory SD within 2× the A-S bound on clean data (thresholds set in Phase 1) before adversarial co-training begins.

Limitations. Several design choices bound the scope of conclusions. First, the market-replay assumption means the adversary does not receive fills on spoofed orders and injected depth does not move prices endogenously; this underestimates adversary impact in thin books and overestimates it in deep ones. Second, single-instrument evaluation (AMZN 2024) limits out-of-distribution generalisability; the out-of-sample phase uses a held-out date range on the same instrument, not a different stock or asset class. Third, oracle detection labels are unavailable in production, so reported AUROC figures cannot be directly translated into a live detection rate. Fourth, the binary regime indicator is a coarse proxy for the continuous spectrum of liquidity regimes; it will not capture intraday regime transitions or the gradual onset of volatility events. These limitations are acknowledged explicitly in the thesis and inform the scope of any practical deployment claims.

5 Timeline, Resources, and Supervision

Phase	Period	Key deliverables
1	Feb–Mar	JaxMARL-HFT baselines (AMZN 2024); adversary cost calibration (SEC/DOJ)
2	Apr–May	Regime indicator validation; progression gate
3	Jun–Jul	IPPO co-training; detection head; regime integration (2-week Aug buffer)
4	Aug–Sep	Ablation; out-of-sample validation; draft due mid-September
5	Oct–Nov	Submission, defence, code release

Data. AMZN 2024 LOBSTER (primary, matching Mohl et al. [3]); DataBento or Dr. Wen's group data (contingency). **Compute.** ≈1,500–2,000 A100-equivalent GPU-hours, Pawsey Supercomputing Centre (pending; contingency: UWA School of Physics, Mathematics and Computing cluster).

Supervision. Dr. **Ajmal Mian** (primary): detection head architecture, causal masking, evaluation under distributional shift. Dr. **Yuanji Wen** (co-supervisor): simulation realism, adversary cost calibration. Dr. **Garrison Gao** (external advisor): adversary design and financial realism. Bi-weekly meetings throughout; weekly during Phases 3–4.

6 Ethical Considerations

All experiments use historical data in closed simulation; no models will be deployed in live markets. The adversary design will be released under UWA's open-access policy with two mitigations: the economic cost calibration limits uplift beyond what real spoofing practitioners already know, and a time-limited thesis embargo may be applied in consultation with supervisors to allow defensive tools to reach exchanges and regulators before full methodology is public. The project follows ACM publication ethics guidelines and will undergo UWA ethics committee review.

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